Proof. First remove the definable locus where the output RV-tuple is not locally constant in the input VF-coordinates. On each resulting open fiber with both input and output RV-tuples fixed, extend the VF-map by zero outside its domain and apply [17, Corollary 9.9] to its partial derivatives. Their nonexistence and discontinuity loci are definable and have VF-dimension less than n in every such fiber. Retaining or projecting away RV-coordinates does not increase VF-dimension, so the combined exceptional locus still has VF-dimension less than n. Off this locus all partial derivatives, and hence their determinant, are continuous. In arity zero the Jacobian is the empty determinant 1. □

Resolved referee R035 [Gap] – Equal VF dimensions and generic RV constancy.
Original source line 993. State $n = n'$ in Lemma 2.38; otherwise the determinant called the Jacobian has not been defined. Moreover, the preceding existence of one full-dimensional fiber for each input t does not show that such fibers cover almost every input point. Use the standard generic local constancy of a definable map from VF^n to $\mathrm{RV}^{m'}$ first, and then apply the cited differentiability theorem on its open fibers. This provides the missing almost-everywhere coverage.

Author resolution. The author approves the square-dimension hypothesis $n = n'$ and the generic local-constancy argument for the output RV-labels. The proof now applies the cited partial-differentiability theorem on open fixed-label charts and explains the smaller-dimensional exceptional locus. The arity-zero Jacobian is the empty determinant 1.

Terminology 2.39. We say that a set $I \subseteq \Gamma_\infty^n$ is *γ-bounded*, where $\gamma \in \Gamma$, if it is contained in the box $[\gamma, \infty]^n$, and is *doubly γ-bounded* if it is contained in the box $[-\gamma, \gamma]^n$. More generally, let A be a subset of $\mathrm{VF}^n \times \mathrm{RV}_\infty^m \times \Gamma_\infty^l$ and

$$\mathrm{valim}(A) = \{(\mathrm{val}(a), \mathrm{vrv}(t), \gamma) \mid (a, t, \gamma) \in A\} \subseteq \Gamma_\infty^{n+m+l},$$

where $\mathrm{val}(a)$ and $\mathrm{vrv}(t)$ are taken coordinatewise. We say that A is *γ-bounded* if $\mathrm{valim}(A)$ is, and likewise for *doubly γ-bounded*. A set is *bounded* if it is γ-bounded for some $\gamma \in \Gamma$, and *doubly bounded* if it is doubly γ-bounded for some $\gamma \in \Gamma$. The notation A_Γ continues to mean projection to existing Γ-coordinates.

Resolved referee R036 [Clarification] – A valuation image is not a coordinate projection.
Original source line 1001. The notation A_Γ here conflicts with Notation 2.1, where it denotes projection to existing Γ-coordinates. Use a separate notation such as $\mathrm{valim}(A)$ for this coordinatewise valuation image. Also explicitly say that 'bounded' means γ-bounded for some γ, and similarly for 'doubly bounded'.

Author resolution. The author approves $\mathrm{valim}(A)$ for the coordinatewise valuation image, reserving A_Γ for projection to existing Γ-coordinates. The existential finite bound in "bounded" and "doubly bounded" is now explicit. This coordinatewise convention is distinct from the minimum tuple valuation used in differentiation.

Definition 2.40 (Contractions)**.** A function $f : A \longrightarrow B$ is rv-*contractible* if there is a (necessarily unique) function $f_\downarrow : \mathrm{rv}(A) \longrightarrow \mathrm{rv}(B)$, called the rv-*contraction* of f, such that

$$(\mathrm{rv} \upharpoonright B) \circ f = f_\downarrow \circ (\mathrm{rv} \upharpoonright A).$$

Similarly, it is res-*contractible* (respectively, val-*contractible*) if the same holds in terms of res (respectively, val or vrv, depending on the coordinates) instead of rv.

Lemma 2.41. *Let $U \subseteq \mathrm{RV}^k$ be a doubly bounded set and $f : U \longrightarrow \mathrm{RV}$ a definable function. Then $f(U)$ is also doubly bounded. The same holds if the codomain of f is Γ.*